LITERATURE REVIEW

Consistent Individualized Feature Attribution for Tree Ensembles

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ANALYTICAL ABSTRACT

The paper asks how a prediction of a gradient boosted or random forest model should be divided among its input features, and shows that the three attribution rules in common use for trees (gain, split count and the path based method of Saabas) are inconsistent: a model can be modified so that it depends more on a feature while that feature receives less credit. The authors argue that SHAP values, Shapley values of the conditional expectation game, are the only additive attribution satisfying local accuracy, missingness and consistency, and they supply Tree SHAP, a recursive algorithm that computes them exactly in $O(TLD^2)$ time instead of $O(TL2^M)$. They extend the construction to pairwise SHAP interaction values and propose summary plots, dependence plots and clustering on attributions. Evidence comes from a 34 person survey on a two feature tree, runtime comparisons, clustering on census and Alzheimer data, a perturbation test on airline tweets and a survival model on NHANES I. The reviewer verifies the central counterexample by hand and finds it correct, but notes that the exactness holds only for the path dependent value function, whose treatment of correlated features is not examined.

1 Introduction

1.1 Paper identification and classification

The paper is a methodological contribution in computational statistics for machine learning, in the subfield of model interpretation and cooperative game theory applied to additive tree models. It combines an axiomatic argument, an algorithm with a complexity result, and five small empirical demonstrations. The version read in full is arXiv version 3 (March 2019), obtained as complete text including all algorithms, equations, captions and the reference list. The figures themselves were not available as images in the text extraction, so statements about them rely on the captions and the surrounding text, and the reviewer reconstructs the central example independently below.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

The authors seek an attribution rule for tree ensembles that is simultaneously faithful to the model, comparable across features and cheap enough to run on whole datasets. The finding they pursue is that the existing tree specific rules fail an elementary monotonicity requirement and that a repaired rule exists in closed form.

WHY DOES THE QUESTION MATTER?

Attributions are used to select features, to audit models for accountability, to debug, and, increasingly, to produce reasons for individual decisions in credit, medicine and insurance. If the rule can assign less credit to a feature whose true influence has grown, then every ranking of features built on it is unreliable, and a regulator or a portfolio manager reading such a ranking is misled in a way that no sample size can repair.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Gain (Breiman et al., 1984), split counts and permutation importance were global tools. The only individualized tree method was Saabas (2014). Model agnostic Shapley based explanations existed (Štrumbelj and Kononenko, 2014; Lundberg and Lee, 2017) but required sampling or exponential enumeration. Missing were a proof that tree rules are inconsistent and a polynomial exact algorithm for the Shapley construction on trees.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes Tree SHAP and SHAP interaction values. It argues by counterexample and axiom, then evaluates by a user study, runtime, clustering explained variance, a feature perturbation test and illustrations. SHAP wins every comparison reported. The reviewer judges the theory and complexity

result to be the durable contribution and the empirical comparisons to be illustrative rather than probative.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Can feature attributions for a prediction of a tree ensemble be defined so that they sum to the prediction minus the baseline, assign nothing to unobserved features, never decrease when a model leans more on a feature, and still be computed in time polynomial in the tree size?

2.2 Hypotheses

- Gain, split count and the Saabas rule violate consistency, so they can reverse the true ordering of two features.
- SHAP values are the unique additive attribution that satisfies local accuracy, missingness and consistency for the conditional expectation value function.
- The exponential cost of Equation (2) can be avoided for trees by tracking the proportion of feature subsets of each size that reach each leaf.
- Attributions that are consistent produce better clusters, better identification of influential features and more intuitive credit assignment.

2.3 Existing literature and research gap

PAPER The authors place the work between the Shapley value literature in cooperative game theory, the local explanation literature (Ribeiro et al., 2016) and tree importance measures. The gap is that tree specific individualized attributions lacked an axiomatic basis and model agnostic ones lacked speed.

CRITIQUE The literature on tree importance is treated briefly, and the discussion of how the conditional expectation value function behaves when features are correlated is absent. That omission matters, as discussed in the critical evaluation.

2.4 Claimed contribution

CRITIQUE Three contributions are claimed: the inconsistency result, the algorithm and the interaction values. The inconsistency result is demonstrated by one two feature example that the reviewer verifies. The algorithm is specified in pseudocode with a stated complexity but no proof of correctness appears in the paper. The interaction values are defined but their axiomatic properties

are only asserted to follow from similar axioms. The contributions are real, but the demonstration is thinner than the confidence of the abstract suggests.

3 Theoretical Framework and Mechanism

3.1 Additive attribution and the axioms

PAPER An additive feature attribution method explains a prediction through a linear function of binary indicators of feature presence.

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i, \qquad z' \in \{0,1\}^M,\ \phi_i \in \mathbb{R}. \tag{1}$$

M	Number of input features.
z'_i	Equals one when feature i is treated as observed, zero when unknown.
ϕ_0	Baseline, the expected model output when no feature is observed.
ϕ_i	Attribution of feature i to the prediction for the instance x.

Equation (1) fixes the form of the explanation but not the values. Three requirements pin the values down. Local accuracy requires $g(\mathbf{1}) = f(x)$, so that the attributions add up to the prediction. Missingness requires $\phi_i = 0$ whenever $z'_i = 0$. Consistency requires that if a model f' satisfies $f'_x(S \cup \{i\}) - f'_x(S) \geq f_x(S \cup \{i\}) - f_x(S)$ for all subsets S, then $\phi_i(f') \geq \phi_i(f)$. The paper takes from Lundberg and Lee (2017) that a unique solution exists once a value function is chosen.

3.2 The value function and the SHAP formula

PAPER The value function evaluates the model on a subset of observed features by conditional expectation, and the attribution is the classical Shapley value of that game.

$$f_x(S) = E[f(x) \mid x_S], \qquad \phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \Big[f_x(S \cup \{i\}) - f_x(S) \Big]. \tag{2}$$

N	Set of all M features.
S	A subset of features treated as observed.
$f_x(S)$	Expected prediction when only the features in S are fixed at their values in x.
Weight	The probability that S is exactly the set of features preceding i in a uniformly random ordering of N.

The weight in Equation (2) is the key to its meaning. A random ordering places i at position $|S| + 1$ with probability $1/M$, and, given that, the preceding set is a specific S with probability $1/\binom{M-1}{|S|}$. The product equals $|S|!(M - |S| - 1)!/M!$, so the formula is an average of marginal contributions over all

$M!$ orderings:

$$\phi_i = \frac{1}{M!} \sum_{\pi} \left[f_x(P_i^{\pi} \cup \{i\}) - f_x(P_i^{\pi}) \right], \tag{3}$$

where P_i^{π} is the set of features preceding i in ordering π. Telescoping the marginal contributions along any one ordering gives $\sum_i [f_x(P_i \cup \{i\}) - f_x(P_i)] = f_x(N) - f_x(\emptyset) = f(x) - E[f(X)]$, and averaging orderings preserves this identity, which is local accuracy. The Saabas rule is exactly one such ordering, the order of splits along the decision path, and that is the origin of its inconsistency: a single ordering depends on tree structure rather than only on the function.

3.3 Worked reconstruction of the inconsistency (reviewer calculation)

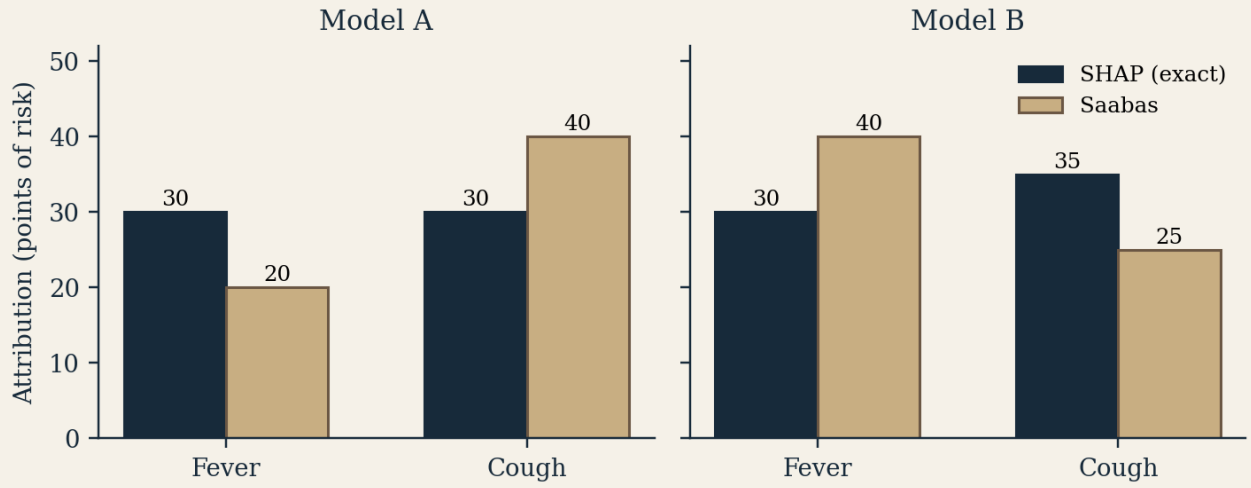
EVIDENCE The paper's Figure 1 uses two trees on Fever and Cough, each of the four leaves holding equal data. Model A is the AND function with output 80 when both are present and zero otherwise, so the baseline is 20. Model B adds 10 when only Cough is present, giving leaves $90, 10, 0, 0$ and baseline 25. The paper states the baselines and the predictions for the instance Fever and Cough both yes, and says Cough is split first in Model B. The reviewer reconstructs the remaining leaf values from that description and verifies the paper's statements.

For Model A, $f_x(\emptyset) = 20, f_x(\{F\}) = f_x(\{C\}) = 40, f_x(\{F, C\}) = 80$. Both orderings give each feature 20 then 40, so $\phi_F = \phi_C = 30$. The Saabas rule splits on Fever first and gives $\phi_F = 40 - 20 = 20$, $\phi_C = 80 - 40 = 40$. For Model B, $f_x(\emptyset) = 25, f_x(\{C\}) = 50, f_x(\{F\}) = 45, f_x(\{F, C\}) = 90$. Averaging the orderings,

$$\phi_C = \tfrac{1}{2}\big[(50 - 25) + (90 - 45)\big] = 35, \qquad \phi_F = \tfrac{1}{2}\big[(90 - 50) + (45 - 25)\big] = 30, \tag{4}$$

which sum to $65 = 90 - 25$. Saabas splits on Cough first and returns $\phi_C = 25$, $\phi_F = 40$. Cough matters more in Model B than in Model A at every coalition, yet Saabas lowers its credit from 40 to 25 and ranks it below Fever, while exact SHAP raises it from 30 to 35. Gain shows the same failure under the equal leaf assumption: in Model A the total squared deviation is 4800, the Fever split removes 1600 and the Cough split 3200; in Model B the total is 5700, the Cough split removes 2500 and the Fever split 3200. Cough gain falls from 3200 to 2500 and is ranked below Fever. Figure 1 reproduces these numbers.

Figure 1. **Individual attributions for the instance Fever and Cough both present, in points of risk**



Source: generated by the reviewer from the leaf values reconstructed from the description of Figure 1 in the paper; exact SHAP computed by enumeration of the two orderings; Saabas computed along the stated split order.

3.4 Tree SHAP: from exponential to polynomial

PAPER For a single tree, $E[f(x) \mid x_S]$ is computed recursively. At a node splitting on a feature in S the path follows x; at a node splitting on a feature outside S the value is the cover weighted average of the two children:

$$G(j, w) = \begin{cases} w\, v_j & \text{if } j \text{ is a leaf}, \\ G(a_j, w) \text{ if } x_{d_j} \le t_j \text{ else } G(b_j, w) & \text{if } d_j \in S, \\ G\left(a_j,\ w\, \frac{r_{a_j}}{r_j}\right) + G\left(b_j,\ w\, \frac{r_{b_j}}{r_j}\right) & \text{if } d_j \notin S. \end{cases} \tag{5}$$

v_j	Output stored at leaf j.
a_j, b_j	Left and right children of node j.
t_j, d_j	Threshold and splitting feature of node j.
r_j	Cover of node j, the number of training samples reaching it.

Evaluating Equation (5) for each of the 2^M subsets inside Equation (2) costs $O(TL2^M)$ for T trees with at most L leaves. Algorithm 2 avoids the enumeration by carrying, down each path, a vector m whose entries hold a feature index d, a fraction z of subsets that exclude the feature and still reach the node, a fraction o that include it and reach the node, and a weight w for each subset size. Following the pseudocode, EXTEND appends a feature with fractions (p_z, p_o) and updates, for the existing path of length l,

$$m_{i+1}.w \leftarrow m_{i+1}.w + p_o\, m_i.w\, \frac{i}{l}, \qquad m_i.w \leftarrow p_z\, m_i.w\, \frac{l-i}{l}, \qquad i = l-1, \dots, 1, \tag{6}$$

and UNWIND inverts this recursion, for a given path entry i, by running $j = l-1, \dots, 1$ with $t = m_j.w$, $m_j.w = n\, l/(j\, m_i.o)$ and $n = t - m_j.w\, m_i.z\, (l-j)/l$ when $m_i.o \neq 0$, and $m_j.w = m_j.w\, l/(m_i.z\, (l-j))$ otherwise.

At a leaf j the contribution to feature $m_i.d$ is

$$\phi_{m_i.d} \leftarrow \phi_{m_i.d} + w_i \, (m_i.o - m_i.z) \, v_j, \qquad w_i = \sum \mathrm{UNWIND}(m, i).w. \tag{7}$$

The factor $(o - z)$ is the difference in the share of subsets that include versus exclude the feature at that leaf, weighted by the Shapley coefficient of the subset size. If a feature repeats on a path, its earlier entry is unwound and its fractions multiplied, which reproduces the conditional expectation semantics of Equation (5). Per leaf the path has at most D entries and each unwind costs $O(D)$, giving $O(D^2)$ per leaf and $O(TLD^2)$ in total with $O(D^2+M)$ memory. For a sum of trees the values add, because Shapley values are linear in the value function.

CRITIQUE The paper presents Algorithm 2 without a proof that the weights w equal the Shapley coefficients $|S|!(l-|S|-1)!/l!$ at each leaf. The reviewer checked the recursion against the two feature example above by hand and the extension step reproduces the coefficients for path length two; a general proof by induction on path length is straightforward but is left to the reader.

3.5 SHAP interaction values

PAPER Pairwise effects use the Shapley interaction index (Fujimoto et al., 2006):

$$\Phi_{i,j} = \sum_{S \subseteq \mathcal{N} \setminus \{i,j\}} \frac{|S|! \, (M - |S| - 2)!}{2(M-1)!} \, \nabla_{ij}(S), \quad \nabla_{ij}(S) = f_x(S \cup \{i,j\}) - f_x(S \cup \{i\}) - f_x(S \cup \{j\}) + f_x(S), \tag{8}$$

and the main effect is the residual

$$\Phi_{i,i} = \phi_i - \sum_{j \neq i} \Phi_{i,j}. \tag{9}$$

The second difference ∇_{ij} is zero for any additive function, so $\Phi_{i,j}$ isolates departures from additivity. Because the bracket in the second line of the paper's Equation (5) equals the SHAP value of i with j present minus the SHAP value of i with j absent, two runs of Algorithm 2 per feature give the whole matrix in $O(TMLD^2)$. By construction $\sum_j \Phi_{i,j} = \phi_i$, so row sums reproduce SHAP values and the double sum reproduces the prediction minus baseline.

4 Data and Empirical Strategy

4.1 Data

PAPER Five small evaluations are reported. None is a controlled benchmark in the econometric sense, and the paper has no sample selection discussion.

Table 1. **Datasets and models used in the paper**

Application	Data	Model	Sample used
Human intuition	Two feature tree (Model A)	None; participants allocate 60 points	34 Mechanical Turk participants
Runtime	Synthetic, features and depth grown together	XGBoost, 50 trees; also 1,000 trees, depth 10, 100 features	One input vector explained
Supervised clustering, census	UCI census income	XGBoost, 500 trees, depth 6, shrinkage 0.005	2,000 individuals explained
Supervised clustering, Alzheimer	200 gene expression modules, CERAD scores	Not stated in the text	Not stated
Influential features	Airline tweets, 1,686 bag of words features, score from -1 to 1	XGBoost, 50 trees, depth 30	11,712 training, 2,928 test tweets
SHAP plots	NHANES I with follow up, 14 features	XGBoost Cox model, 7,000 trees, depth 3, shrinkage 0.001, 50 percent subsampling	9,932 individuals, 20 year follow up, 80/20 split for tuning

Source: copied from Sections 5.1 to 5.5 of the paper (official). Entries marked not stated are not reported in the text read by the reviewer.

4.2 Empirical specification and identification

CRITIQUE The paper does not estimate a parameter, so classical identification does not apply. The empirical question is whether an attribution method recovers a ground truth influence. Ground truth exists only in the two feature example, where the AND structure makes equal credit the natural answer, and in the perturbation test, where the proxy for truth is the change in the model output when a feature is replaced by a random training value. This proxy resembles the interventional value function and not the conditional one used to define SHAP, which slightly favours methods that behave like permutation. The comparison is therefore internally coherent but the identifying variation, the random replacement, is chosen by the authors.

4.3 Data limitations and biases

- Model hyperparameters are reported but variability across training seeds is not, so differences of 0.01 in area under the curve (census, 0.98 against 0.97) cannot be interpreted.
- Survey participants were not screened for expertise and no demographic or order effect information is given, and the number of participants is small.
- The tweet test reports one summed curve per method, with no confidence band.
- Hyperparameters of the NHANES model were chosen by an 80/20 split and the final model retrained on all individuals, so the summary and dependence plots describe an in sample fit.
- No multiple testing issue arises, as no hypothesis tests are reported; the corresponding weakness is the absence of any measure of uncertainty.

5 Results

5.1 Agreement with human intuition

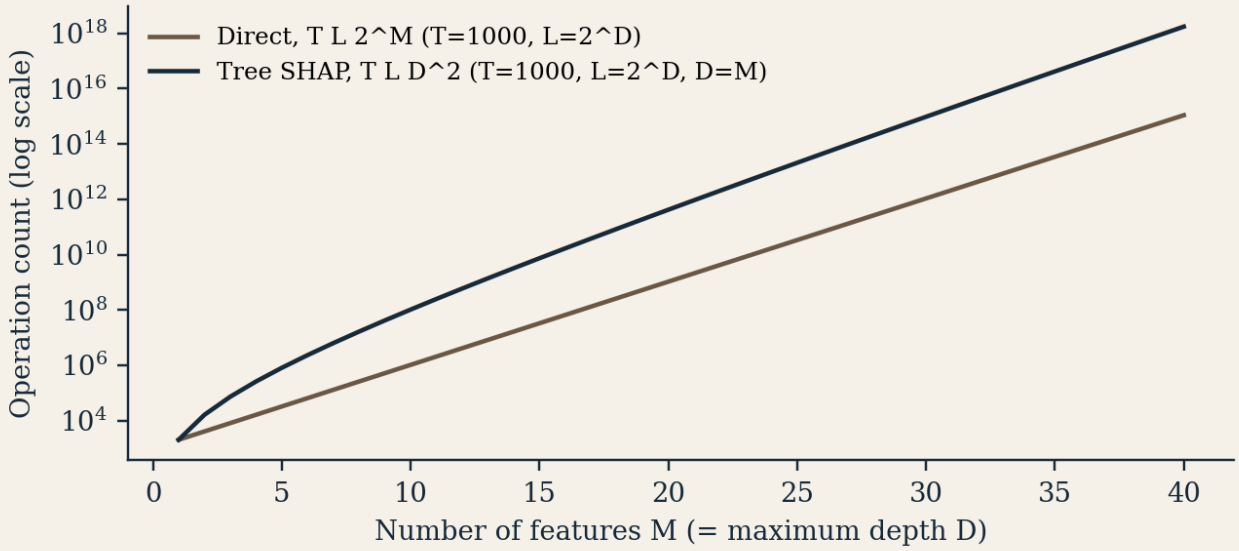
EVIDENCE Thirty four participants split the 60 point increase in risk of Model A between Fever and Cough. The paper reports that the equal split of 30 and 30 was the most common allocation, a smaller group favoured the first split feature, and fewer still followed the Saabas allocation. Exact counts are shown only in a figure whose values the text does not give. With 34 responses the standard error of a share near one half is about 0.086 ($\sqrt{0.25/34}$), so a plurality claim is credible but any statement about the relative size of the minority groups is not.

CRITIQUE The example is symmetric by design, so a preference for equal credit is the answer any symmetry axiom would give. The study tests agreement with symmetry, not agreement with Shapley weighting in asymmetric structures.

5.2 Computational performance

EVIDENCE An XGBoost model with $1{,}000$ trees of depth 10 over 100 features is explained in 0.08 seconds. Direct enumeration at $M = 100$ would require $2^{100} \approx 1.3 \times 10^{30}$ subsets per tree, which is infeasible, so the speedup is a change of complexity class and not a constant factor. Figure 2 computes the two operation counts for a stylised case; it is an analytical illustration of the stated complexities and not a timing.

Figure 2. **Operation counts of the direct and the polynomial algorithm as features and depth grow together**



Source: generated by the reviewer from the stated complexities $O(TL2^M)$ and $O(TLD^2)$, with $T = 1{,}000$, $L = 2^D$ and $D = M$; constants ignored, not measured timings.

5.3 Supervised clustering

EVIDENCE Hierarchical clustering is run on attributions instead of raw features, and quality is the share R^2 of model output variance explained by group means as groups are merged. The paper reports an area under that curve of 0.98 for SHAP against 0.97 for Saabas on the census model, and 0.96 against 0.88 for the Alzheimer model. The census gap is one percentage point on an area bounded by one, and with no variance estimate it is not demonstrably different from zero. The Alzheimer gap of 0.08 is larger and more informative. The result means that grouping individuals by their reasons for a prediction separates the predicted outputs well, which is close to a tautology because R^2 is computed on the model output itself and attributions sum to that output.

5.4 Identification of influential features

EVIDENCE For each of $2{,}928$ test tweets the most negative feature identified by each method is replaced with its value from a random training tweet, and the cumulative increase in predicted sentiment is plotted. SHAP is reported to yield the largest total increase. Global methods can select only one feature for the whole dataset, so the authors replace it only when it would likely raise the score. This concession treats gain and split count generously, but the comparison is still weighted against them by construction, as a single feature cannot adapt to individual tweets.

5.5 SHAP plots on NHANES I

EVIDENCE The summary plot (sorted by $\sum_j |\phi_i^{(j)}|$) shows age as the dominant mortality risk, systolic blood pressure influential for a minority with high values, and long right tails but short left tails for several measurements. The dependence plot shows that a given blood pressure has a larger risk contribution at younger ages, and interaction values attribute most of the vertical dispersion in that plot to the age by pressure term. In a Cox model the output is a log relative hazard, so an attribution of 0.5 to a feature multiplies the hazard by $e^{0.5} \approx 1.65$, holding the baseline fixed. The paper does not report magnitudes on this scale.

CRITIQUE The medical interpretations (a high pressure being less alarming with age) are labelled presumptive by the authors, and they describe model behaviour and not physiological mechanism.

6 Robustness and Additional Tests

Table 2. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Counterexample is special to one tree	None beyond Figure 1	Gain, split count and Saabas fail	Partly. A counterexample suffices to refute consistency, but it gives no sense of frequency in trained models
Intuition result depends on symmetry	One tree, 34 participants	Equal split most common	No. An asymmetric tree is not tested
Clustering metric favours attributions that sum to output	Two datasets	SHAP above Saabas	Partly. Both methods sum to the output, so the comparison is fair, but no variance is reported
Perturbation measure favours interventional methods	Random replacement from training data	SHAP above four rivals	Partly. Replacement is not the conditional expectation that defines SHAP
Hyperparameter sensitivity	None	Not reported	No
Dependent features	None	Not reported	No

Source: reviewer assessment of the experiments described in Sections 2 and 5 of the paper.

CRITIQUE The paper contains no robustness section in the usual sense. Each experiment is a single configuration, and the table above records what each can and cannot support.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Gain, split count and the Saabas method are inconsistent in the sense defined, shown by an explicit example the reviewer verified. Tree SHAP computes the conditional expectation Shapley values exactly in polynomial time, which follows from the stated algorithm and the linearity of Shapley values across trees.

SUPPORTED BUT CONDITIONAL

That SHAP values are the unique consistent and locally accurate attributions holds under the conditional expectation value function and the additive form of Equation (1); the uniqueness proof is inherited from Lundberg and Lee (2017), not given here.

SUGGESTIVE

That supervised clustering and SHAP plots improve understanding of models rests on one illustration each and on area under curve differences of unknown precision.

NOT ESTABLISHED

That SHAP identifies the features that truly drive the outcome in the world, or that attributions are stable when features are correlated, is not addressed. Attributions explain a fitted model, not the data generating process.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Exponential speed improvement, which makes whole dataset explanations practical and motivates the new plots.
- A theoretical guarantee (consistency) that rival methods lack.
- Implementation merged into XGBoost and LightGBM, which supports replicability.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The counterexample is minimal and checkable by hand, which makes the inconsistency claim unusually transparent.

- Linearity of Shapley values lets exactness extend from single trees to boosted and bagged ensembles without further assumptions, so the result applies to the models practitioners actually use.
- Row sums of the interaction matrix equal the SHAP values, so the decomposition is conservative: nothing is lost or double counted.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The value function $f_x(S) = E[f(x) \mid x_S]$ is evaluated through the tree's own cover, so the conditional distribution of the omitted features is whatever the training paths imply. **Why it matters.** If feature k is not used by the model but is strongly correlated with feature i that is used, the tree computation can still give k no credit while giving i all of it, and for features that substitute for each other along different branches, credit can change with the arbitrary choice of which one the tree split on. A user who reads the attribution as how much the real outcome depends on a feature will over or under state it. **Potential bias.** Attributions depend on training data composition and on the fitted structure, and they can assign nonzero credit to features that are irrelevant to the real process (through correlation) or to features the model uses only because the data contain a spurious relation. **How it could be tested.** Simulate correlated Gaussian features with a known additive truth, fit boosted trees, and compare path dependent Tree SHAP against interventional values and against the known truth over a grid of correlation; later work (Janzing et al., 2020; Sundararajan and Najmi, 2020) addresses exactly this choice.

OTHER WEAKNESSES

- The consistency argument depends on the chosen value function. With a different notion of missingness the uniqueness result gives different numbers, so claims that SHAP is the only valid attribution are conditional (Kumar et al., 2020).
- Attributions are single numbers with no uncertainty, though the model, the baseline distribution and the training seed are all random. Stability under retraining is untested.
- The user study cannot discriminate Shapley weighting from a symmetry heuristic.
- The perturbation test compares one method class with others under an intervention that resembles one of the methods.
- Multiple modelling choices (tree depth up to 30 in the tweet model) are not varied, though deep trees produce many repeated feature paths where the unwinding logic is most delicate.

8 Contradictions and Edge Cases

CRITIQUE Several conditions can be examined directly from the formulas.

INFERENCE First, duplicated features. Suppose features 1 and 2 are identical copies and every split in the model uses feature 1 only. Feature 2 then never appears on a decision path, so its marginal contribution is zero in every coalition and $\phi_2 = 0$, whereas ϕ_1 receives all the credit. A human reading would regard both copies as equally responsible. The missingness property, applied to the structure of the fitted model, thereby produces an arbitrary split of credit that depends on which copy the learner happened to choose.

INFERENCE Second, the baseline. Local accuracy ties the sum to $f(x) - E[f(X)]$, so attributions describe a deviation from the average prediction, not from a neutral reference. In Model B above the baseline is 25 and not zero; for an instance with a prediction near the average, all attributions are small even if the model relies heavily on features whose effects cancel. For a population shifted away from training, the baseline moves and attributions change although the instance and model do not.

INFERENCE Third, scale. In the exponential regime 2^M the speedup is dramatic, but the polynomial bound has D^2, and for deep trees ($D = 30$ in the tweet model) the path vector can hold up to thirty entries, so the per leaf cost grows by nearly three orders of magnitude relative to depth five. The ratio $30^2/5^2 = 36$ is a hand calculation from the stated bound, not a measured timing.

INFERENCE Fourth, interaction values for higher order structure. With three features and a parity style model, pairwise terms capture only part of the structure; the matrix is a projection of a three way effect onto pairs, so the interpretation of a zero pairwise value as no interaction is incorrect in that case.

CRITIQUE Fifth, ranking reversals under retraining. Two correlated features of near equal usefulness can swap order between seeds, and consistency holds only for a fixed trained model. The theory says nothing about the variance of attributions between two models trained on the same data.

9 Further Literature

Table 3. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Shapley (1953)	Value of a cooperative game via average marginal contribution	Foundation	Supplies Equations (2) and (3)
Strumbelj and Kononenko (2014)	Shapley values to explain model predictions	Preceded; model agnostic and sampled	Establishes the use of Shapley values; Tree SHAP removes sampling
Lundberg and Lee (2017)	Additive attributions, uniqueness theorem	Theoretical base	Gives the axioms the present paper imports

Paper	Shared idea	Relationship	Why it matters
Saabas (2014)	Path based attribution for trees	Criticised target	The single ordering rule shown inconsistent
Fujimoto et al. (2006)	Shapley interaction index	Foundation	Underlies Equation (8)
Janzing et al. (2020)	Interventional versus observational value functions	Contradicts the conditional choice as a causal measure	Shows that attributions under correlation need not reflect causal influence
Sundararajan and Najmi (2020)	Many Shapley methods	Qualifies uniqueness claim	Uniqueness depends on the value function
Kumar et al. (2020)	Problems with Shapley based importance	Critique	Lists conceptual issues with reading attributions as importance
Lundberg et al. (2020)	Tree SHAP in applications, interventional version	Extended and applied	Adds a background data version and medical applications
Gu, Kelly and Xiu (2020)	Tree and neural models for asset returns	Applied setting	Interpretation of return predictors by importance measures

Source: reviewer's classification of the cited works; relationships are the reviewer's judgement.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** behaviour under correlated features. *Why it matters:* real tabular data are correlated. *Existing limitation:* the path dependent value function. *Way forward:* compare against interventional and causal Shapley variants on simulated truth.
- **Gap:** sampling uncertainty of attributions. *Why it matters:* rankings used for decisions need error bars. *Existing limitation:* point values only. *Way forward:* bootstrap retraining and report rank stability.

10.2 Literature-level gaps

- **Gap:** link from model attribution to economic or causal effects. *Why it matters:* attributions are often misread as effects. *Existing limitation:* attributions describe the fitted function. *Way forward:* combine with experimental or quasi experimental variation.
- **Gap:** a standard benchmark with known ground truth attributions. *Why it matters:* claims of improvement are currently unfalsifiable. *Existing limitation:* ad hoc datasets. *Way forward:* synthetic generators with controlled correlation and interaction.

10.3 Methodological gaps

– **Gap:** higher order interactions beyond pairs. *Why it matters:* parity and threshold structures. *Existing limitation:* cost $O(TMLD^2)$ for pairs only. *Way forward:* efficient higher order Shapley interaction indices for trees.

11 Research Opportunities

Idea 1. Attribution stability in return predictability models

RESEARCH QUESTION. How stable are Tree SHAP rankings of return predictors across retraining windows and seeds?

MOTIVATION. Characteristic importance rankings in empirical asset pricing are used to argue which signals matter.

HYPOTHESIS. Rank correlation of mean absolute attributions across seeds falls sharply for correlated characteristics.

ECONOMIC MECHANISM. Substitutable predictors share credit arbitrarily across trees.

LITERATURE CONNECTION. Gu, Kelly and Xiu (2020); Kumar et al. (2020).

DATA. CRSP and Compustat firm characteristics, monthly, as in public replication sets; availability to be verified.

VARIABLES. Monthly returns, around ninety characteristics, mean absolute SHAP per characteristic.

METHOD. Rolling boosted tree training with many seeds, Kendall rank correlation of attributions.

IDENTIFICATION STRATEGY. Variation in seeds and windows holding the data fixed.

EXPECTED CONTRIBUTION. A measure of reliability for importance claims.

MAIN RISK. Data licences and compute cost.

Idea 2. Correlation stress test with known truth

RESEARCH QUESTION. When do path dependent and interventional Tree SHAP differ materially?

MOTIVATION. The most important weakness above.

HYPOTHESIS. Divergence grows monotonically in feature correlation and in tree depth.

ECONOMIC MECHANISM. Cover weighting encodes the training dependence among features.

LITERATURE CONNECTION. Janzing et al. (2020); Sundararajan and Najmi (2020).

DATA. Simulated; fully generated.

VARIABLES. Correlation, depth, number of trees, attribution error against truth.

METHOD. Monte Carlo design with closed form true attributions for additive and interacting models.

IDENTIFICATION STRATEGY. Controlled simulation.

EXPECTED CONTRIBUTION. A map of the regions where each variant is reliable.

MAIN RISK. Results may be known to specialists but not collected in one design.

Idea 3. Attribution clustering for regime detection

RESEARCH QUESTION. Do clusters of SHAP vectors from a volatility forecasting model identify regimes better than clusters of raw inputs?

MOTIVATION. Raw features have mixed units; attributions share the units of the forecast.

HYPOTHESIS. Attribution clusters explain more out of sample variance of realised volatility.

ECONOMIC MECHANISM. Regimes are defined by which drivers matter, not by feature levels.

LITERATURE CONNECTION. Present paper, Section 5.3.

DATA. Daily index returns and realised volatility from public sources; availability to be verified.

VARIABLES. Lagged realised measures, implied volatility, SHAP vectors.

METHOD. Hierarchical clustering and the R^2 merge curve of Figure 6 of the paper.

IDENTIFICATION STRATEGY. Out of sample comparison to raw feature clustering.

EXPECTED CONTRIBUTION. A test of the unit free property claimed by the authors.

MAIN RISK. Clusters may reflect the forecast level only.

Idea 4. Pairwise interaction values for credit scoring audits

RESEARCH QUESTION. Do interaction values reveal proxies for protected attributes that main effects hide?

MOTIVATION. Regulatory audits require detecting indirect discrimination.

HYPOTHESIS. Proxy variables show large interactions with income variables.

ECONOMIC MECHANISM. Interaction between a proxy and legitimate variables transmits protected information.

LITERATURE CONNECTION. Equation (8).

DATA. Public mortgage disclosure data; availability to be verified.

VARIABLES. Loan outcome, borrower covariates, interaction matrix.

METHOD. Boosted trees with Tree SHAP interaction values.

IDENTIFICATION STRATEGY. Comparison across subgroups with matched covariates.

EXPECTED CONTRIBUTION. A diagnostic with an explicit decomposition.

MAIN RISK. Legal and measurement difficulties.

Idea 5. Uncertainty intervals for attributions

RESEARCH QUESTION. Can resampling provide valid confidence intervals for mean absolute SHAP?

MOTIVATION. Point values hide sampling variation.

HYPOTHESIS. Naive bootstrap intervals undercover for deep ensembles.

ECONOMIC MECHANISM. Not applicable; statistical.

LITERATURE CONNECTION. Gap on uncertainty above.

DATA. Simulated and public benchmark sets.

VARIABLES. Coverage, interval width.

METHOD. Bootstrap, subsampling, and jackknife after bootstrap.

IDENTIFICATION STRATEGY. Coverage on simulated truth.

EXPECTED CONTRIBUTION. Calibration of attribution intervals.

MAIN RISK. Cost of repeated training.

11.1 Ranking

Table 4. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Stability in return models	4	3	3	4	4	4	22
2. Correlation stress test	3	5	5	5	4	3	25
3. Clustering for regimes	3	4	4	3	3	3	20
4. Interactions in credit audits	4	2	2	3	4	4	19
5. Attribution intervals	4	4	5	4	4	4	25

Source: reviewer's subjective scoring.

The top three are Ideas 2 and 5 (tied) and Idea 1. Ideas 2 and 5 are feasible with generated data and answer the principal weakness directly. Idea 1 ranks third because it transfers the diagnostics to a setting of economic interest where the data are costly.

12 Learning Framework

12.1 Prerequisite concepts

- Decision trees, gradient boosting and cover.
- Cooperative games, the Shapley value and its axioms.
- Conditional expectation and the difference between conditioning and intervening.

12.2 Core concepts learned

- Consistency as a monotonicity requirement for attributions.
- Local accuracy as an adding up constraint on the baseline.
- Tracking subset sizes allows a Shapley sum over 2^M terms to collapse.

12.3 Important equations

Equation (2) defines the attribution, Equation (3) gives its ordering interpretation, Equation (5) the tree conditional expectation, Equation (6) the EXTEND recursion and Equation (8) the interaction index.

12.4 Methods learned

Recursive evaluation of conditional expectations on trees, path vectors carrying subset size weights, and attribution based clustering.

12.5 Questions to test understanding

COMPREHENSION

1. Why do the two orderings in Model B give Cough 35 and Fever 30?
2. What does local accuracy require of the sum of attributions?

ANALYSIS

1. Why must the algorithm track each subset size rather than the total number of subsets?
2. Why is the interaction matrix symmetric by construction?

RESEARCH

1. How would a causal value function change the counterexample of two correlated features?
2. What test would show that an attribution ranking is unstable?

12.6 Papers to read next

Lundberg and Lee (2017); Janzing et al. (2020); Lundberg et al. (2020); Kumar et al. (2020).

13 Conclusion

13.1 One-sentence thesis

Tree attribution methods based on a single split order are inconsistent, and the Shapley value of the conditional expectation game can be computed exactly for trees in polynomial time.

13.2 Three most important insights

- A single ordering of features cannot be a fair attribution for a nonlinear model; averaging over orderings is required.
- Shapley values are linear across trees, so exact computation scales to ensembles.
- Attributions explain a fitted model relative to a baseline and depend on the value function used.

13.3 Strongest evidence

The two feature counterexample, verified here, with exact numbers for Saabas and gain.

13.4 Biggest weakness

Dependence on the path dependent conditional expectation when features are correlated.

13.5 Most important research gap

A benchmark with known ground truth attributions and reported uncertainty.

13.6 Best research opportunity

A simulation design measuring divergence between path dependent and interventional attributions as correlation rises.

13.7 Three papers to read next

Lundberg and Lee (2017); Janzing et al. (2020); Sundararajan and Najmi (2020).

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